



A Watermark for Large Language Models

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Why Watermark?

Problem: LLM-generated text is hard to distinguish from human writing. As LLM's become indistinguishable from many human texts, they can be used to create harmful content like:

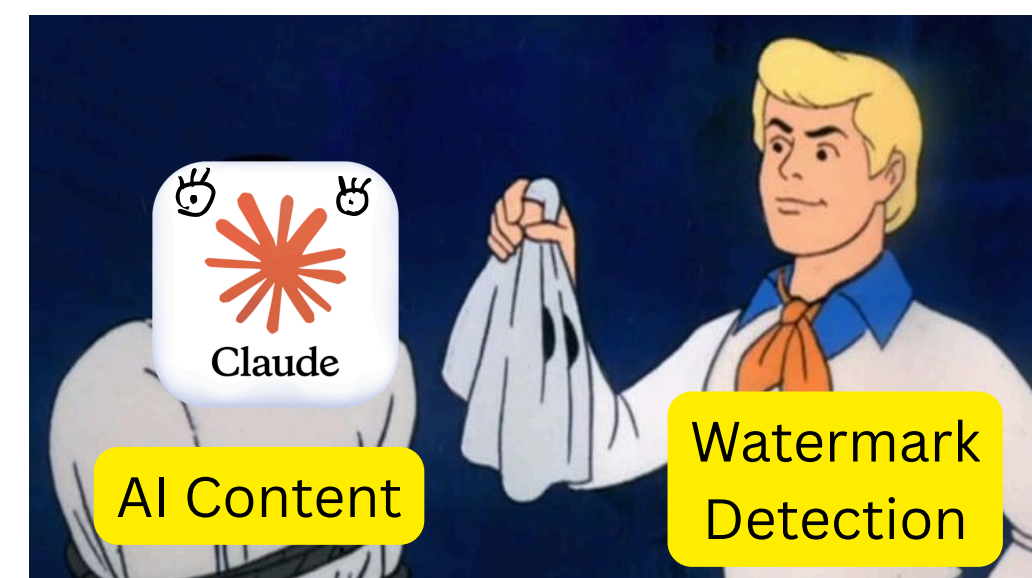
Spam messaging

Cheating

Synthetic Data

Plagiarism

Fake News



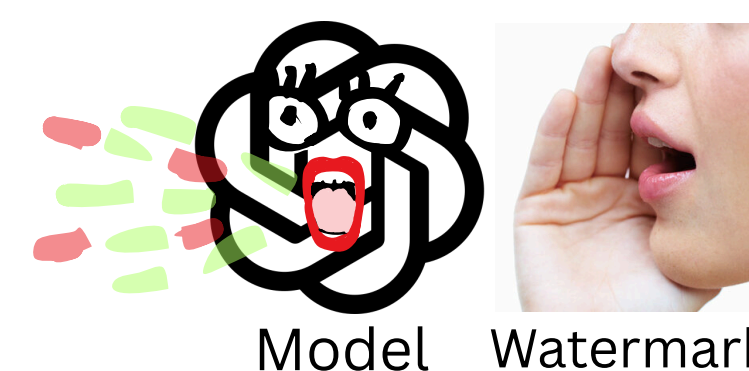
Watermarking Wishlist

Hard to Dismantle 🤔

BEFORE
This essay –
was written
entirely – by
an AI – robot

AFTER
This paper was,
certainly and
entirely, written
by me, and not an
AI robot.

Doesn't Modify Weights



Efficient
generation

Easy to verify

Negligible Quality
Reduction

High Confidence
Detection

One Algorithm, Different Generators

Alex's
hash_key

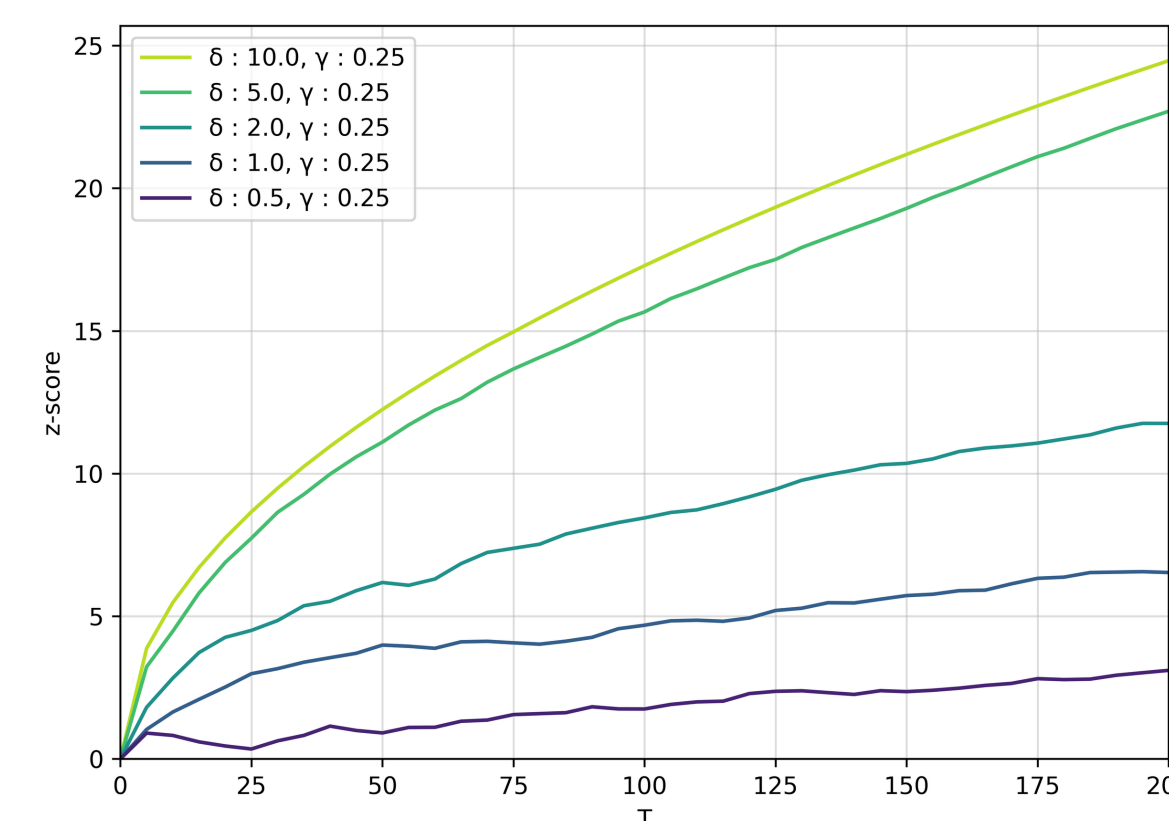
Zach's
hash_key

Public OpenAI
hash_key

Who generated this text?
I like cats and dogs...
I like cats and dogs...
I like cats and dogs...
I like cats and dogs...

$p < 0.05$ $p > 0.05$ $p > 0.05$

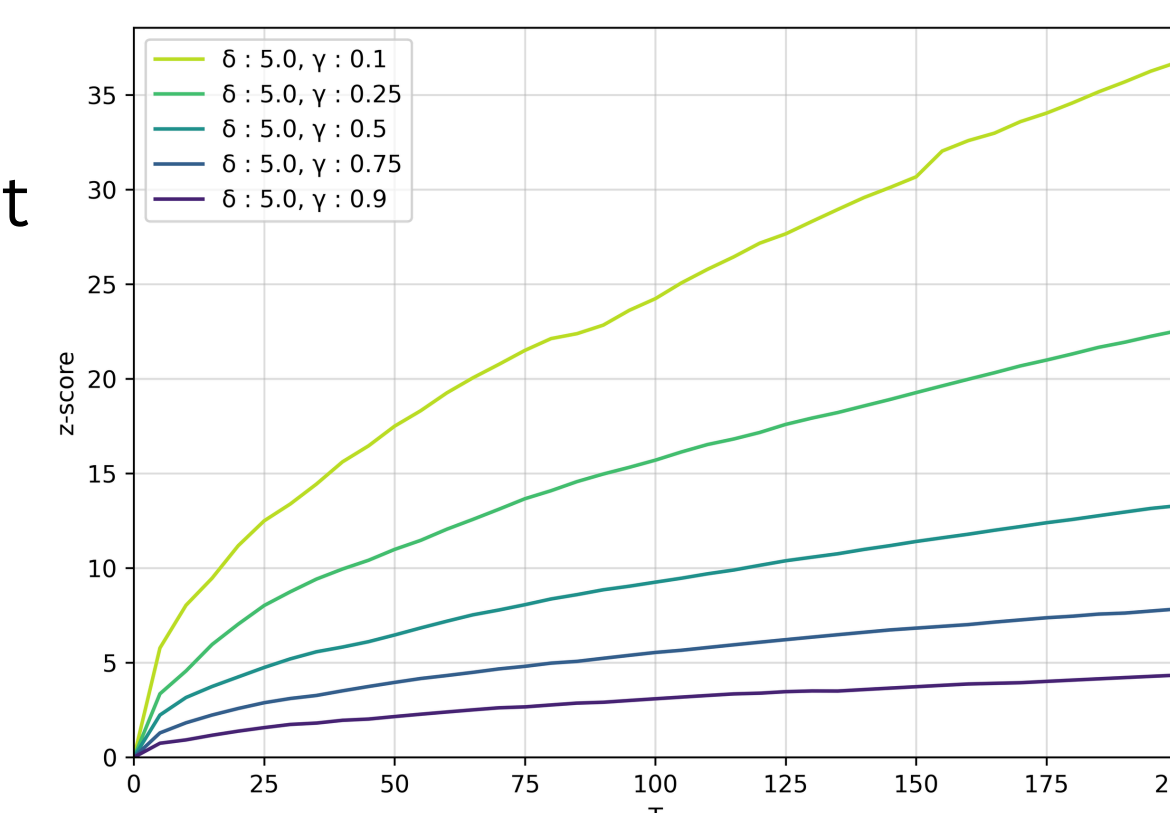
Z-Score Analysis



- Average z-score increases with number of tokens generated
- Steepest increase is early, showing it doesn't require a large amount of tokens for a high z-score
- Tradeoff: larger δ means higher z-scores, but less natural language
- Tradeoff: smaller γ means higher z-scores, but less natural language

Error Rate Analysis

		z = 4.0		z = 5.0	
δ	γ	FPR	TPR	FPR	TPR
1.0	0.25	0.000	0.956	0.000	0.941
1.0	0.5	0.024	0.795	0.000	0.706
2.0	0.25	0.000	1.000	0.000	1.000
2.0	0.5	0.000	1.000	0.000	1.000
5.0	0.25	0.024	1.000	0.000	1.000
5.0	0.5	0.000	1.000	0.000	1.000



- True positive rate increases as delta increases
- True positive rate decreases as the z-threshold increases, but false positives become rarer
- At most one false positive was observed for any (δ, γ) pair
- Re-running with a larger sample size will give more accurate data

Models & Data

- We used OPT-1.3B for our analysis above, like the original paper.
- Prompts were manually selected or sampled from the c4 dataset.
- For our original watermark detection game, we used meta-llama/Llama-3.2-3B-Instruct.

Conclusions

- Watermarking has inherent trade-off between detectability and quality of generated text
- We still find it successful: strong detection rate and text is not obviously watermarked
- Issue: still susceptible to attacks

Further Research?

- **Attacks:** Can an adversarial model rephrase watermarked text to remove the signal without humans noticing
- **Human Detection Game:** With more participants, and a larger model, does the mean approach 50%?

Methodology

Setup

gamma: Influence
how many tokens
are green γ

delta: How strongly
the model favors
green tokens δ

hash_key: Secret
number that privatizes
your random seed K

Generate

THE DOG LIKES...

BONES 2.7
WALKS 2.5
...
READING -2.6

BONES 2.7
WALKS 2.5 + δ
...
READING -2.6 + δ

L
O
O
P

THE DOG
LIKES
WALKS
...

Sample next token

Detect

THE DOG LIKES WALKS ...

THE DOG LIKES...
WALKS, BALL, READING, ...
BONES, PARK, LIBRARY, ...

Green / n tokens

$p = 0.0005$

$E[\#G] = n \cdot \gamma$

Game Results

- About 250 rounds of data, 150 with default params, and 100 with max params
- Default params ($\gamma=0.5$, $\delta=2$) had a 59.3% correctness rate
- Max params ($\gamma=0.9$, $\delta=5$) had an 86% correctness rate

Can You Detect a Watermark?



Try Our Game!

<https://8845662f7a2df668e2.gradio.live>

References

- Kirchenbauer, J., Geiping, J., Wen, Y., Katz, J., Miers, I., & Goldstein, T. (2023). A watermark for large language models. Proceedings of the 40th International Conference on Machine Learning (ICML).